

Functional Enhancement of a Bio-Inspired Flying Fish Robotic Model Organism

George Kopf

April 29, 2026

Advisor: Aimy Wissa

Second Reader: Hossein Valavi

Class: ECE399

Date: April 27, 2026



Submitted in partial fulfillment of the requirements for the degree of
Bachelor of Science in Engineering
Department of Electrical and Computer Engineering
Princeton University

I hereby declare that this Independent Work report represents
my own work in accordance with University regulations.

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Abstract

Princeton's Bio-Inspired Adaptive Morphology (BAM) Lab has an ongoing project using a robot model organism (RMO) to investigate the physical principles behind taxiing behaviors in flying fish. The model is very useful, but is limited by a rigid tether and the lack of a control system. The purpose of this independent work was to integrate electrical design elements to improve the functionality of the RMO as a bio-relevant tool via untethering. A better biological model will allow us to better understand natural physics with improved experiments to inform new technologies. Additionally, the microcontroller-based system will act as a platform for integrating smart systems in future development. In this project, we were able to successfully integrate electrical control and demonstrate tether-less swimming at higher DoF. The RMO has wireless connectivity and internal positional sensing, allowing accelerometer and gyroscope data to be exported from the fish onto a computer remotely teleoperating the RMO, where motor speed and sensor configurations can be controlled.

Acknowledgments

Special thanks to the wonderful people of the BAM Lab including Professor Aimy Wissa and Derrick Roseman, second reader Professor Hossein Valavi, and the ECE & MAE Departments.

Thanks to the Department of ECE and SEAS for providing critical funding, it is deeply appreciated.

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Figure 1: *Exocoetidae* flying fish, photo credit Neil Bowman

1 Introduction

The *Exocoetidae* family of "flying fish" is famous for its members' fantastical ability to leap from the waves and glide vast distances traveling over 30 km/h and hundreds of meters in distance [1], [2]. This unique class of fish manages to do what no other fish can by swimming aggressively towards the surface of the water, lifting all but the ventral side of their caudal fin out of the water as they spread their pectoral wings and engage in taxiing. As flying fish fully leave the water and glide, they take advantage of acute ground effect forces and wind dynamics along the surface waves, increasing lift and reducing drag. In a good wind, flying fish can extend their flight from 50 meters to over 400 [3]. These fish engage in complex and powerful fluid interactions near the water's surface but are very difficult to study using traditional natural biology methods [4].

In order to better understand the physics at play, researchers have diversified approaches, including in depth simulations [5] and robotic models for physical study [6]. To this, the BAM lab and its partners have developed a series of bio-inspired RMOs to be used as biologically accurate proxies for the flow dynamics of flying fish. Several iterations of the RMO have been developed over the years to study various aspects of the flying fish, from caudal fin dynamics to flexible wing mechanics [6], [7]. All previous versions have heavily focused

on body morphology to study flow dynamics, focusing on anatomical fidelity and using the simplest actuation method possible to generate caudal beating. The original RMO has direct actuation via a DC motor wired to an external power supply. Rigidly tethered, it is excellent for studying head-on flow dynamics but limits experimental design versatility; the original RMO is just a motor driving a rigid shell. It has no way to balance or steer and thus needs to be rigidly attached to a guide rail in order to operate. The RMO only has a single DoF, so it is limited to experiments involving linear movement forwards or at a set angle.

This independent work project set out to improve the tools used in testing. By upgrading the RMO, we can advance our capability to recreate and study the physics that flying fish biology takes advantage of. Better tools lead to more diverse experiments lead to a deeper understanding of both the biological and the engineered. The intent of this project was to improve the RMO's experimental capabilities through the integration of electrical power systems, an internal battery, and a programmable microcontroller.

2 Design Methodology

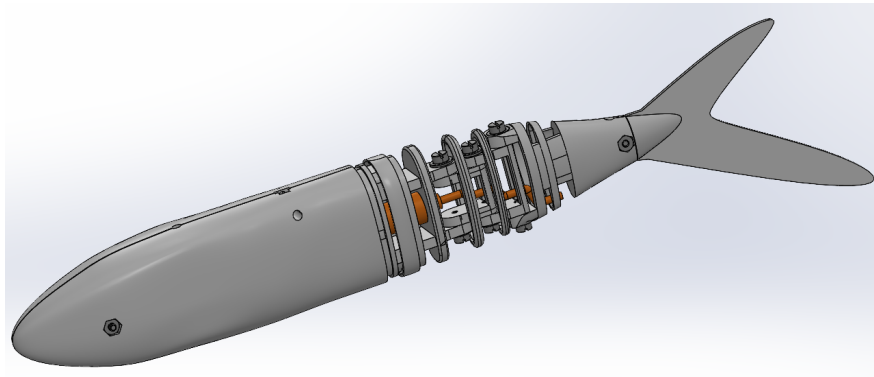


Figure 2: Original RMO CAD file, provided by Derrick Roseman

2.1 Design Goals

As a standard benchmark, the original RMO (see Fig. 2) can swim with a 2Hz swimming gait on 5V with 0.3A of current pull, and can be pushed even further at higher voltages. This project sought to build an electrical power system which could push the motor to similar speeds at 5V (and a current cap of 1A) without needing to be tethered to a bench-top DC power supply. By controlling motor actuation with a programmable microcontroller, the RMO can moderate its own swimming gait and speed, allowing for testing of various caudal fin behaviors. A 6 DoF Inertial Measurement Unit (IMU) allows for gyroscope and

accelerometer data to be recorded by the RMO, reducing the difficult to collect accurate positional and force data. The scope of the independent work settled on this level of control, but the possibility for sensor-based directional steering and balancing becomes accessible once the microcontroller is integrated.

The project focused on not only creating a more accurate model of the fish, but also a usable one! Quality of life and long-term reliability are critical to the model's viability as an experimental tool. Significant time was spent out planning every element to be robust, convenient, functional, and safe, all while minimizing volume and weight. Before the RMO is placed in the water, it must be completely sealed with a coating of silicon epoxy. Once sealed, getting to the electronics inside requires tearing through that coating and resealing the body, which invariably introduces leaks, weak points, or surface deformations which affect flow dynamics. The goal was to make the internal instruments robust and long lasting, sufficient to run many trials at load without needing to pause for disruptive repairs.

As an electrical system, one of the main concerns is safety. The original RMO consistently accrues moisture in the body cavity and when sealed improperly can take on significant water. With just a replaceable motor inside, this isn't that big of a deal. With a lithium-polymer battery, it poses an electrical fire risk.

Electrical systems also pose new challenges that the original RMO did not have to face. The battery has to be rechargeable – it's unacceptable to crack open the RMO to access the battery every time it needs a recharge. There must be an access point somewhere on the surface of the RMO that can connect to a power supply. This poses a leaking risk because the exposed pins must breach the silicon epoxy seal, as well as an experimental concern; exposed pins may affect the surface profile of the RMO, which could affect flow dynamics and invalidate data collected using the device. Feature implementation must improve the RMO's value as a tool for experimental use.

Additionally, there needs to be a way to hard-disconnect the battery from the rest of the circuit. This is important for safety in the event of a short, but it also vastly improves the convenience of the robot. The quiescent current pull from the microcontroller and ICs will slowly but continually drain the battery unless it is physically disconnected from the circuit. When equipment availability can be limited, it's important that the RMO is ready to go when needed. Having to wait and recharge before every testing session would be really inconvenient. However, a typical mechanical switch cannot simply be stuck to the outside of the RMO for the same leaking and profile-disrupting reasons considered for the charging port. We need an alternative.

All of this must be considered without losing sight of the largest constraints: space and weight. The RMO is carefully designed to match the physical form of a real flying fish. This

is part of what makes it a relevant tool to study them: morphological accuracy. It also unfortunately means that we're stuck with the available space in the body cavity and can't change our volume easily to increase buoyancy. The RMO has roughly a 60mm x 40mm x 30mm space inside, in which all our components have to fit without weighing down the RMO.

2.2 Power System

The original RMO only needs to slip a pair of wires from the terminals of the motor up through the body and over to a DC power supply. By removing this tether and incorporating an internal battery, the power system suddenly requires a few extra things.

3.7V single cell Lithium Polymer batteries like the one used in this project are great for their energy density and high discharge capacities. However, they require over-current protection circuitry and a safe recharging method. We ultimately decided to use a TP4056 charger IC. It's a linear, constant-current/constant-voltage chip that allows for LiPo recharging with minimal circuitry needed, with appropriate voltage/current capacities and thermal properties [8]. The TP4056 is a well known standard for single cell battery charging, and should be an appropriate choice for the 500 and 2000mAh 3.7V LiPo batteries the RMO is built to use (it is designed to have interchangeable battery options to accommodate changing weight and space constraints).

Most of the circuitry in the RMO was designed on a custom PCB board to save on weight and space, as opposed to combining multiple breakout boards with equivalent functionality together. However, the HW373 v1.2.1 represented such a perfect module for this application that implementing it on a PCB wouldn't save much space.

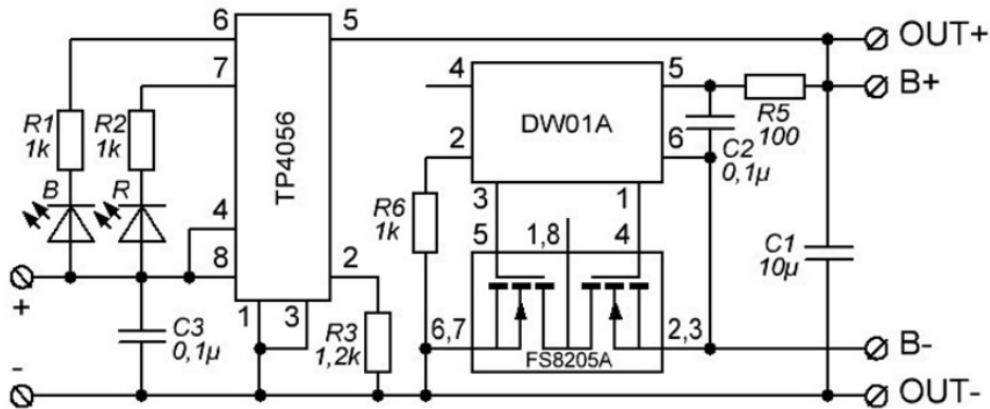


Figure 3: HW373 v1.2.1 schematic [9]

The HW373 breakout board shown in Fig. 3 is perfect for this application. It uses

a TP4056 IC to charge a single cell 3.7V nominal LiPo with LED signaling and charging feedback. The programming resistor, labeled R3 above, forms a voltage divider with an internal resistor in the chip. By replacing this resistor, the charge current can be modified. This is great, because the 500mAh battery and the 2000mAh battery have different charging current requirements. When R3 is $1.2\text{k}\Omega$, current coming out of the TP4056 is 1A, which is appropriate for the 2000mAh battery. As seen in Fig. 4, swapping R3 to $3\text{k}\Omega$ drops capacity down to 0.4A, which is safe for the 500mAh battery. Pushing current above safe levels for a LiPo battery can cause overheating, dendrite formation, and battery failure, none of which are acceptable [10]. Using a complete breakout board allows for the battery & charging module to be installed as a complete unit which can be entirely replaced relatively easily if the battery needs to be swapped. On a side note, the HW373 slowly drops current pull as the battery charges, eventually shutting off when the current drops below 10% of its starting value – this functions as a convenient charge level indicator.

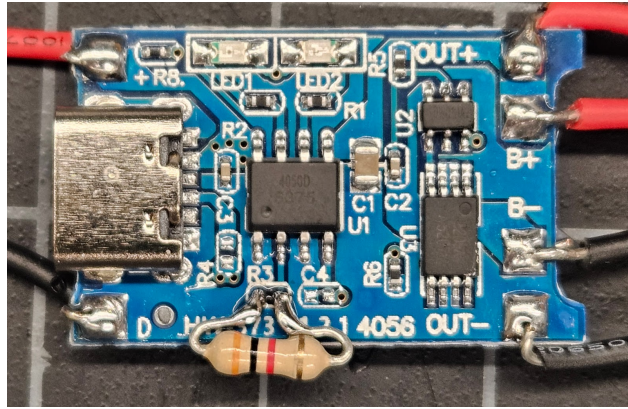


Figure 4: HW373 breakout module with $3\text{k}\Omega$ prog resistor replacement for 500mAh battery

The HW373 offers additional functionality. The main reason it was chosen was due to the inclusion of a DW01A protection IC that works with a FS8205A dual MOSFET. This powerful part of the circuit is a critical safety system. It provides overcharging (4.2V maximum charge), over-discharging (2.5V minimum cutoff), and short-circuit protection (3A cutoff) for the battery, providing critical protection in case circuitry downstream breaks – the worst case scenario in the event of a short is starting a battery fire [11], [12]. The inclusion of the HW373 breakout board is critical to the safety and functionality of the RMO.

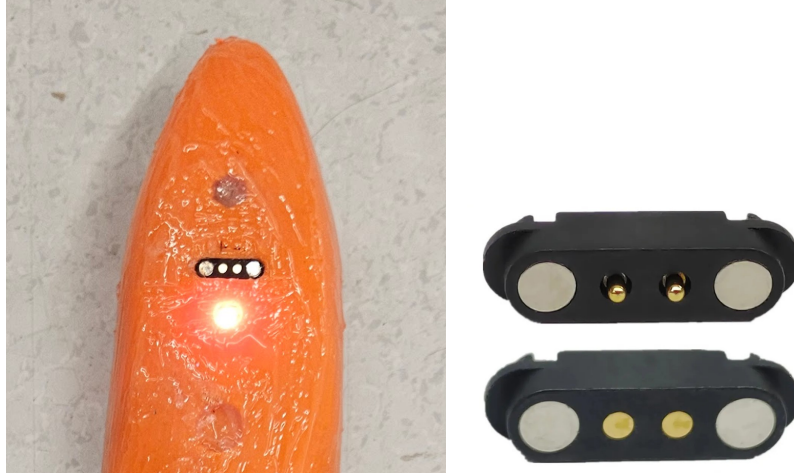


Figure 5: Exposed pogo pin connector on surface of RMO and full pogo pin connectors, photo credit DHgate Electronics

From the charging module, the power-in pins are connected to a magnetic pogo connector piece, as can be seen in Fig. 5. This provides a clean and easy access point to charge the RMO. To avoid leaks around the seams, the pogo pieces were tightly press fit into the body and sealed from the inside with epoxy. The low and flat form factor is intended to mitigate surface profile disruption, although this could be improved further in future versions (the connector mandates a gap in the silicon epoxy coating).

Just implementing the battery protection and charging isn't enough though. As discussed above, we need a way to physically disconnect the battery and charger module from the rest of the circuitry. This is a good safety feature, conserves charge over periods of inactivity, and allows the TP4056 to charge the RMO while the circuit load is disconnected, which improves the chip's thermal properties significantly [9]. We need some kind of switch.

A mechanical switch would poke out of the surface profile and ruin flow dynamics. Waterproof switches and latching buttons are also quite bulky, typically being intended for larger scale purposes. Non-waterproof options cannot be properly sealed with silicon without impeding physical movement. Thus, a magnetic reed switch was chosen. Reed switches are typically simple metal structures that make contact when a magnet pulls them together, resulting in "normally open" behavior. The circuit is open by default and closes when a magnet is introduced. Conversely, normally closed reed switches have the opposite behavior: staying closed by default and opening when a magnet is introduced due to induced reverse polarity pushing connectors apart – typically pushing against a biasing magnet [13]. For the purposes of this project, we used NC reed switches; it is best to not need the magnet to be present while the fish is on and swimming. This way, the magnet can be used as an off trig-

ger. When it is placed against the body of the RMO, the circuit disconnects. An indicator LED was installed across the circuit, to allow for visual confirmation of power state, as can be seen in Fig. 6 below.

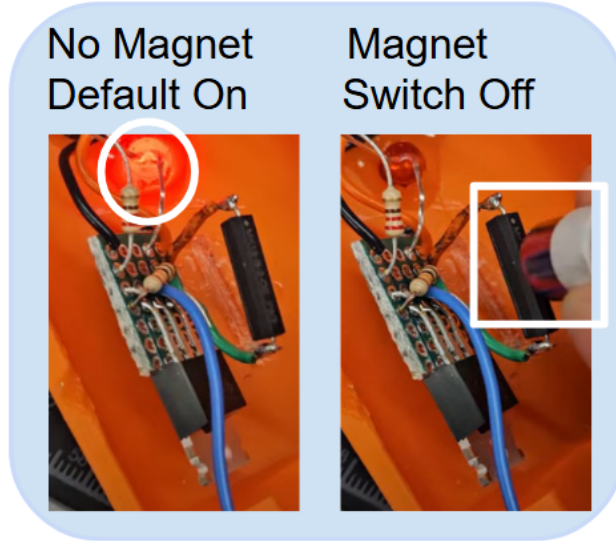


Figure 6: Demonstration of magnetic switch controlling MOSFET low side switching. The LED tracks connectivity between the MOSFET Drain and battery ground, showing that the circuit disconnects when the magnet is introduced.

Small reed switches, such as those utilized in this project, typically feature low current limits, as the thin metal components inside of them will begin to melt above 0.5A [13]. To protect the switch from the full current of the motor, it was instead connected to an IRLZ44N MOSFET [14] via the circuit shown below in Fig. 7. This logic-level MOSFET has a gate threshold voltage at 2-3V, which is well below the nominal 3.7V and maximum 4.2V the battery provides. Thus, our battery can easily switch the MOSFET on with little effective resistance (and thus little waste heat generated). As an n-channel MOSFET, it is better for low side switching, so it is controlling the connection of the ground side.

This design allows for easy power control without compromising the surface of the RMO for flow dynamics or leaks. The reed switch has a detection range of roughly 1cm, which means it can easily sense the magnet through the robot's shell. To maximize ease of use, the magnet was embedded in air dry modeling clay and pressed into place along the body, creating a form-fitting structure that can clip onto the RMO and firmly affix the magnet in place while not in use.

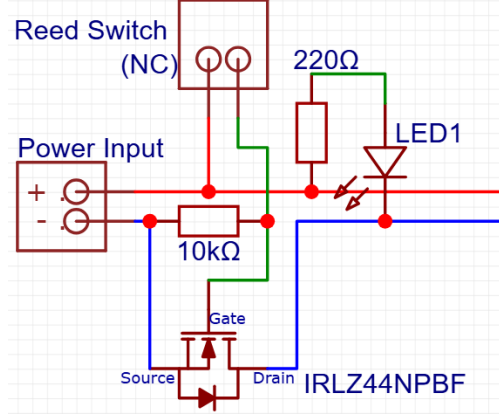


Figure 7: Circuit diagram of implemented NC reed switch. Ground is only connected when the reed switch detects a magnet.

2.3 PCB Design and Implementation

With size and weight being such critical factors, it was determined that building the rest of the circuit into a custom PCB would be a beneficial move. This allows the circuit to be optimized for our specific use case, avoiding wasting space and power efficiency on unnecessary features/general purpose breakout boards. The PCB was designed using easyEDA's standard design platform and purchased from JLCPCB Electronics, and is shown below in Fig. 8

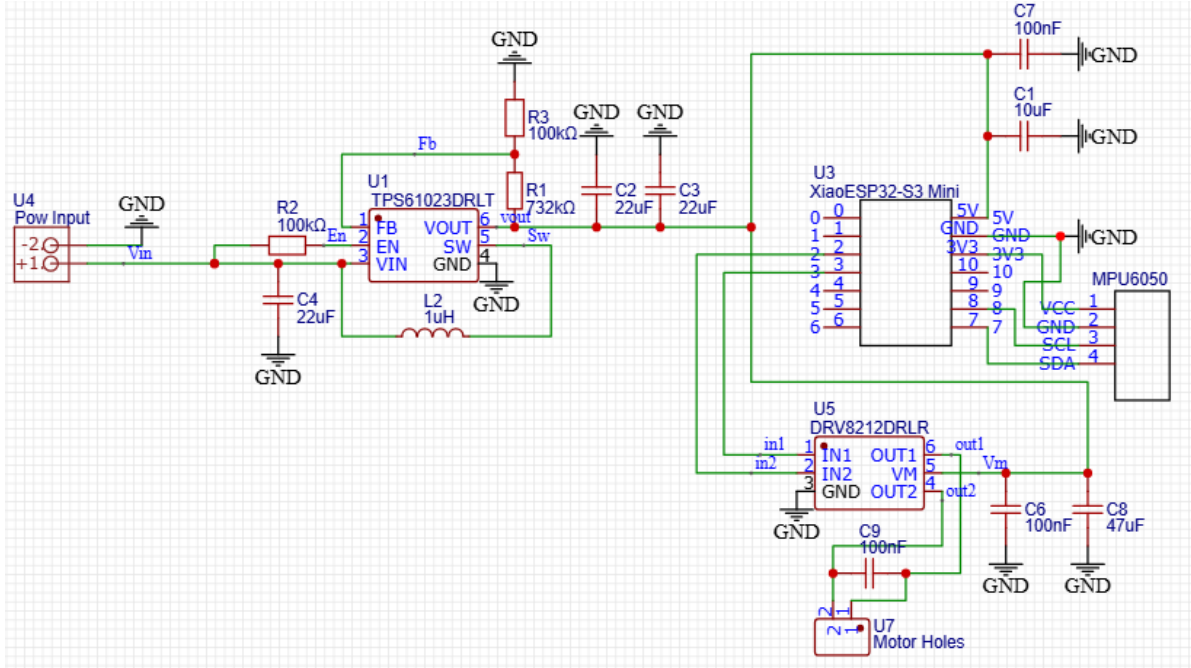


Figure 8: PCB circuit diagram including TPS61023 voltage regulator IC, XIAO ESP32 S3 microcontroller, and DRV8212 motor driver IC.

The voltage regulator is a TPS61023 chip. The TPS61023 is a feedback based synchronous boost converter, easily capable of regulating out battery's variable 4.2 to 3.7V input up to a flat 5V [15]. Using a linear boost converter is not as effective, because voltage will drop as the battery discharges. The TPS61023 has a tunable feedback mechanism that allows for VOut control via a voltage divider. Resistors R1 and R3 set the FB (feedback) pin to a divided value of the VOut pin's output voltage. The chip looks to set FB to 0.5V at all times, meaning it automatically adjusts Vout until it reaches that state. By choosing R1 and R3, VOut can be set to a range of voltages from 0.5V to 5.5V. The setup chosen here sets VOut to 5V exactly. Even as the battery drains and drops from 4.2 to 3.7V and even lower, the TPS61023 will maintain a steady 5V output voltage, which is critical for standardizing the RMO's swimming performances as the battery dies.

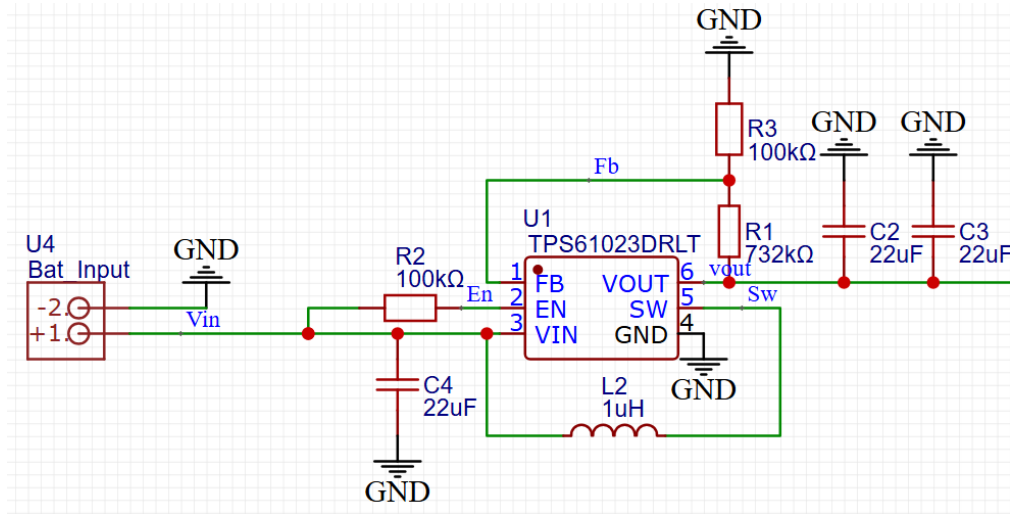


Figure 9: Zoom in on the voltage regulator circuitry.

As can be seen in Fig. 9, input and output capacitors keep the output stable as rapid MOSFET switching pushes variable loads. They dampen oscillatory ringing (voltage ripple) caused by the inductor. The SW (switch) pin rapidly cycles power to inductor L2. The chip pushes power into the inductor, which aggressively resists changing current. This creates a voltage difference that allows the chip to actually do the voltage boosting. When the SW pin cycles off, the inductor can discharge that energy which goes through VOut to the rest of the circuit at 5V. The FB pin affects the cycling behavior, which is how the chip controls VOut.

The TPS61023's feedback loop is very sensitive, and the output capacitors are critical to smooth the signal going back to FB. Fig. 10 shows how the first version of the PCB had a critical mistake which ruined the functionality of the device.

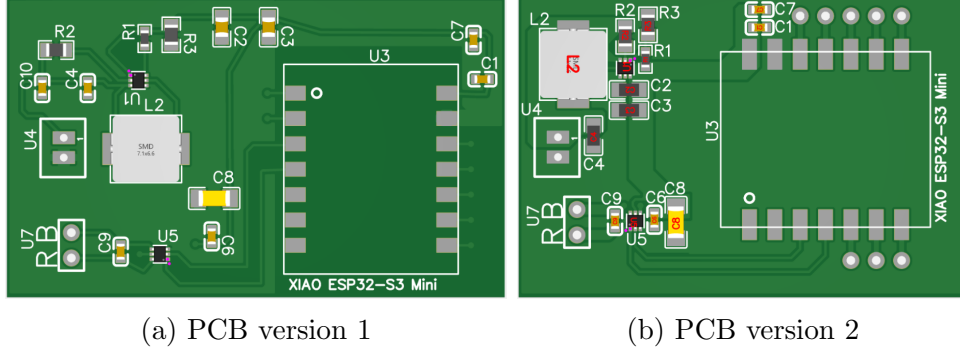


Figure 10: The initial design was flawed because of spread out capacitor placement. Even a few mm can introduce unacceptable levels of impedance, resulting in disastrous voltage ripple.

The TPS61023 really needs output capacitors C2 and C3 and inductor L2 to be as close to the chip as possible. The feedback circuit is so sensitive and the inductor switching circuit is so noisy that they would aggressively leak into each other, causing FB to jump up and down and resulting in messy output signals that did not provide a functional 5V output signal, as can be seen in Fig. 11.

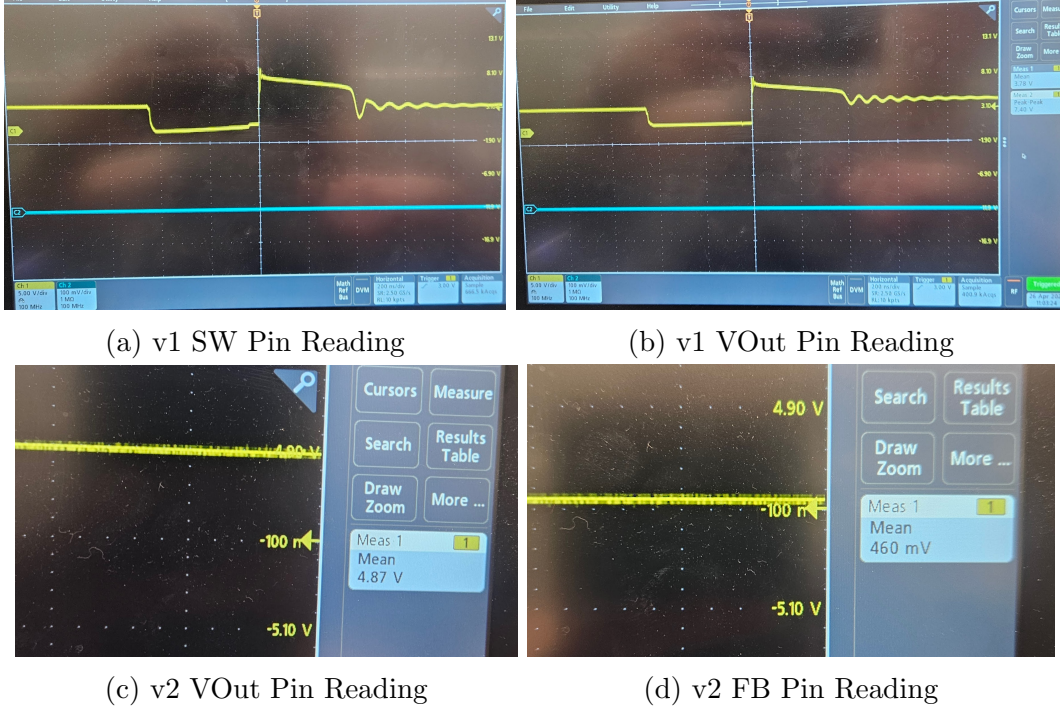


Figure 11: The VOut pin in the initial PCB displays a slightly damped version of the SW pin waveform, indicating the switching noise is strongly leaking into the VOut pin and breaking the circuit logic. This is rectified in the second version.

In the updated PCB, all the capacitors were tightened up, including ones on the motor driver H-bridge chip. This wasn't as strictly mandatory as with the TPS60123 (the version one motor chip did work) but regardless is good practice to establish and helps reduce inefficiencies in the circuitry.

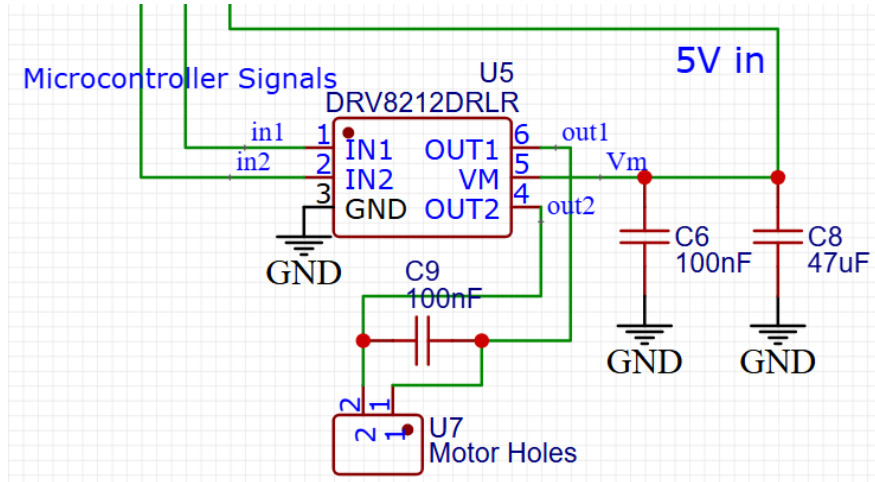


Figure 12: Zoom in on the H-bridge circuitry.

The 5V output from the voltage regulator powers the XIAO ESP32 S3 microcontroller and the DRV8212 N-channel H-bridge motor driver, as can be seen above. The DRV8212 has a 1.65 to 11V operating range with a continuous current capacity of 4A and a peak of 8A [16]. It has built in under-voltage protection to keep MOSFETs from overheating in a partial activation state, over-current protection to protect the circuitry from motor stalls causing current spikes, and an automatic thermal shutdown process to avoid permanent damage when overheating.

As can be seen in Fig. 12, capacitors C6, C8, and C9 provide signal smoothing and bulk capacitance. This helps absorb V_m voltage dips and ringing from motor inductance, protecting the DRV8212 and the rest of the circuit. The DRV8212 doesn't require a flyback diode because it has a built in array of body diodes inside of MOSFETs which handle inductive kickback.

The DRV8212 allows the microcontroller to send PWM signals to In1 and In2, varying motor speed by effectively changing the average voltage being pushed through, and as an H-bridge, allows for motor actuation in both directions. This means the RMO can engage in all sorts of swimming gaits, from staggered beating to alternating directions. This setup could even replicate real-life caudal swimming gaits, such as burst-and-coast [17].

The voltage regulator, motor driver, and microcontroller make up the components of the PCB circuit. Integrating these components on a single board allows for space and power

efficient design, both critical things for the RMO application. Fig. 13 demonstrates the functioning updated version of the PCB.

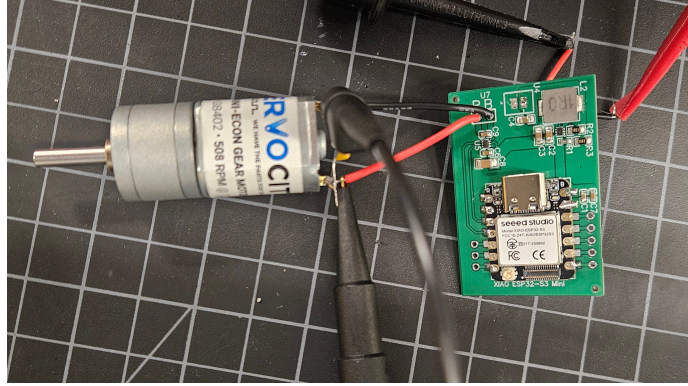


Figure 13: The updated PCB functionally driving the motor via ESP32 control

2.4 Wireless Control & Sensing

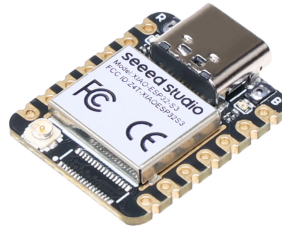


Figure 14: XIAO ESP32 S3 Mini Microcontroller [18]

The XIAO ESP32 S3 Mini was chosen for its small form factor (see Fig. 14 and wireless capabilities. It has an on-chip 8MB PSRAM and a 32GB SD card slot, which is more than enough processing power for our purposes. With built in 2.4GHz WiFi capabilities, it makes wireless connection relatively straightforward to set up [18]. The XIAO is directly soldered onto the PCB using the custom footprint designed using easyEDA. As discussed before, mechanical buttons pose leaking and flow profile issues. Being able to dynamically interact with the RMO's settings remotely is a clean solution to this problem.

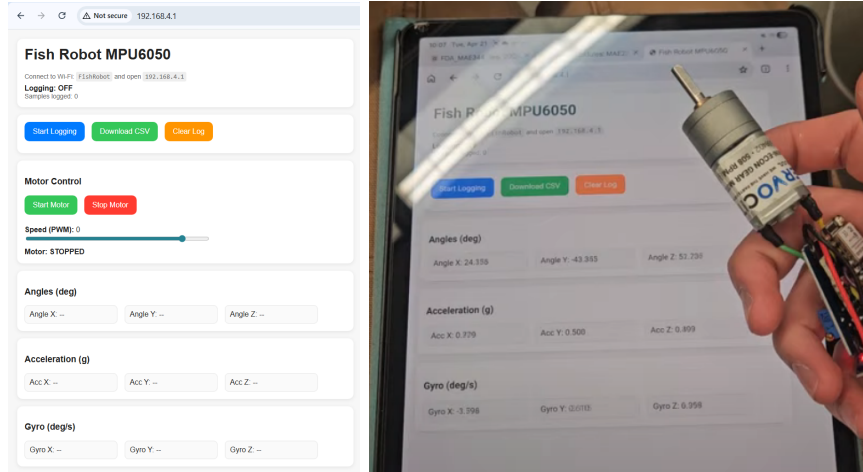


Figure 15: RMO control webpage hosted by the ESP32 microcontroller

By using the `WebServer.h` library, the XIAO can generate its own WiFi network and host a local webpage. HTML code embedded in the Arduino file allows the webpage to load a nice interface with clean buttons and sliders for controlling the RMO, as can be seen in Fig. 15. For this project, the webpage had live PWM motor control and IMU sensor data logging and downloading. IMU accelerometer and gyroscope data can be logged and saved as a .csv file on the visiting device.

The wireless connection plan ran into one major issue; 2.4GHz signals get absorbed by water molecules, rapidly turning into heat. This meant the hosted webpage was functionally useless in the RMO once it hit the water, and any connected device would immediately lose the signal. This was a complete oversight in the design of the system, and there unfortunately was not time to order new parts to overcome this issue. All experimental methods had to work around this issue. In section 4.1, methods to restore connectivity in water will be discussed.

The IMU used for this project was an MPU6050. To measure acceleration, multiple tiny proofmasses are suspended by small springs, one for each axis of measurement [19]. On each side of them are capacitive plates. When acceleration occurs, the masses deflect, changing the capacitance and triggering internal circuitry (analog voltage shifts). An Analog-to-Digital block inside the IMU converts this to a digital signal that represents the amount of deflection. To measure rotation with the gyroscope, each axis contains a vibrating proofmass that is driven by an electrode at a steady state. When rotated, the Coriolis effect changes the vibration patterns, triggering sensing electrodes that record this shift in vibration. Again, an ADC block converts this to a digital signal. By communicating with the XIAO through the SCL clock line and the SDA data line, the MPU6050 sends data on 6 DoF of measurement.

2.5 Waterproofing



Figure 16: The silicone sleeve and silicon epoxy coating work together to seal the body cavity.

It has already been mentioned how critical waterproofing is, due the sensitivity of components to moisture and the danger the LiPo battery poses. A sleeve made of Ecoflex 00-50 silicon snugly fits over the flexible rear structure of the fish, allowing for rapid movement of the resin printed linkages inside without exposing the motor or bearings to the water. Two layers of silicon resin were then carefully used to glue down the sleeve and completely coat the porous PLA body, as is shown in Fig. 16. The original RMO, even when sealed in silicon epoxy, accumulates moisture inside the body cavity through micropores/gaps in the coating. Much attention and care was taken to make the outer seal on this RMO as solid as possible.

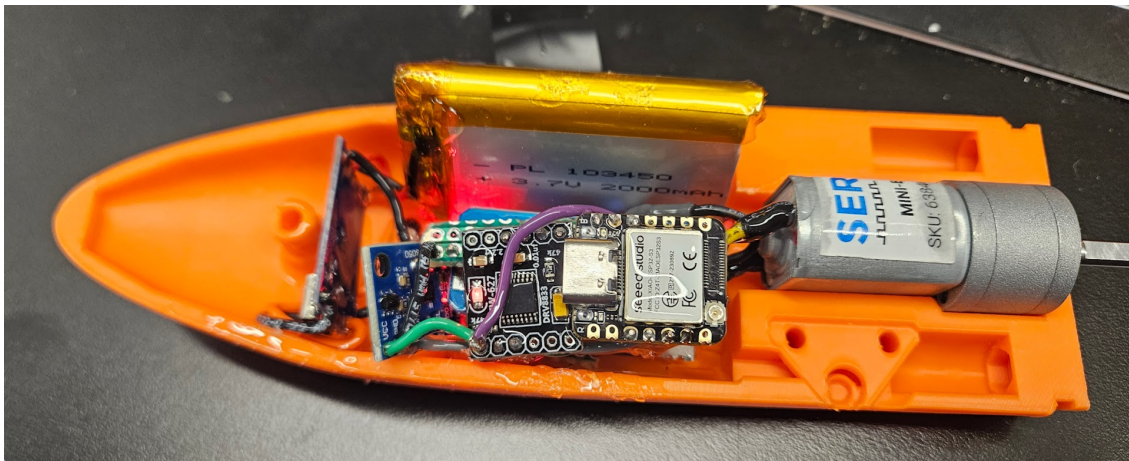


Figure 17: The internal electronics after being painted with conformal coating.

Fig. 17 reveals inside of the body cavity, where all electronics were coated with a layer of conformal coating. This polymer coating helps protect electronics from moisture damage and definitively increases the robustness of the RMO. Conformal coatings have a tradeoff

between protective thickness and thermal insulation, which can result in overheating and cracking from expansion stress. The best is a thin but complete layer that entirely covers the circuitry. Other waterproofing strategies may be investigated in the future, such as filling the body cavity with mineral oil or moving away from a PLA.

2.6 Final Build

The circuitry in Fig. 17 is definitely not a custom PCB. Indeed, due to the time constraints of the project combined with delays in shipping, the critical capacitor placement error in the initial design of the PCB meant that the fixed, second version would not arrive in time for proper performance testing. In order to get the RMO into the water on schedule and prove functionality, a prototype circuit was built using several functionally similar breakout modules that were available to use. The complete circuit is shown below in Fig. 18.

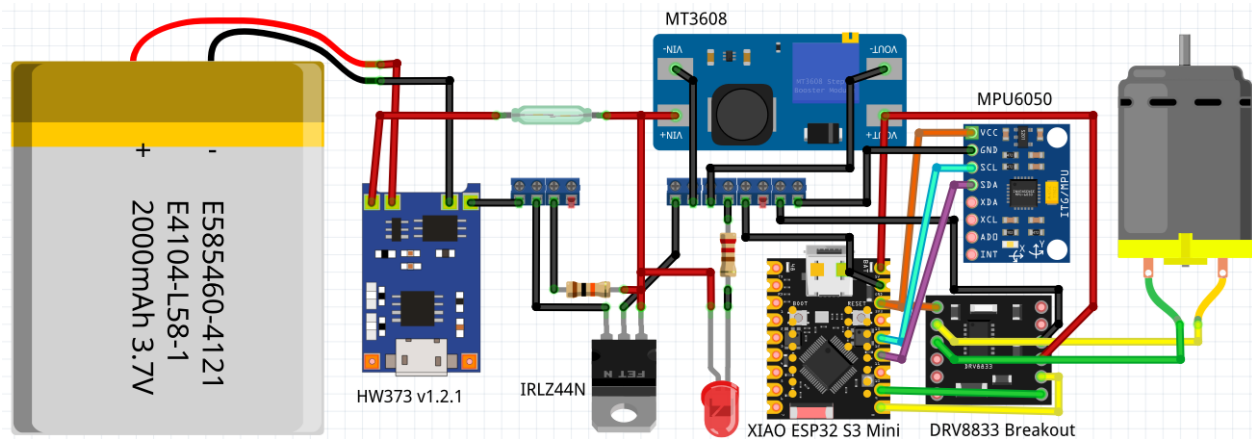


Figure 18: Schematic of breakout module-based circuit used in prototype RMO for functionality demonstration.

Notably, the two main PCB components, the voltage regulator and motor driver, were replaced with similarly functioning breakout modules on hand. The DRV8212 H-bridge motor driver was replaced with a DRV8833 H-bridge motor driver module. The DRV8833 is an older model of the same design, with a smaller voltage range and, importantly, a much lower peak current capacity. The DRV8212 was chosen for its capacity to handle high peak currents up to 8A, which is well above what the motor in the fish should be generating even in stall (the tail does stall sometimes when mechanical components get caught on each other). The DRV8833 chip can handle normal operating conditions perfectly and generally should be fine during motor startup, but with a peak current limit of only 2A, it is at potential risk of failing during stall and definitely limits upper bounds on power [20]. For the purposes of this project's scope, the DRV8833 is an acceptable (but not optimal) component to use for

linear swimming at 5V and less than 1A.

The TPS61023 synchronous voltage regulator is powerful because of its built in feedback mechanism, allowing it to maintain a 5V output over a range of input voltages as the battery depletes over time. The only available breakout module that the lab had which fit the right specs and was the small enough to replace the TPS61023 was an MT3608 DC-DC boost converter (TPS61023 breakout modules do exist but unfortunately we did not have one available). The MT3608 is not powerful. It can well handle our voltage range but, like the DRV8833, also struggles with a 2A current limit [21], [22]. The other big issue with the MT3608 is that it is a linear voltage regulator. There's no feedback control. Instead, the MT3608 has a potentiometer that must be manually tuned to set the output voltage to the desired levels. This means that as the battery discharges, the output voltage of the MT3608 drops as well. For testing, the board was tuned to boost 4.2V to 5.2V, which meant 3.7V was boosted to only 4.6V. Thus, the RMO lost a lot of swimming power as the battery discharged, requiring frequent charging to keep that variable controlled and data consistent. This is problematic for an experimental tool, so it's good that the MT3608 is just a temporary placeholder.

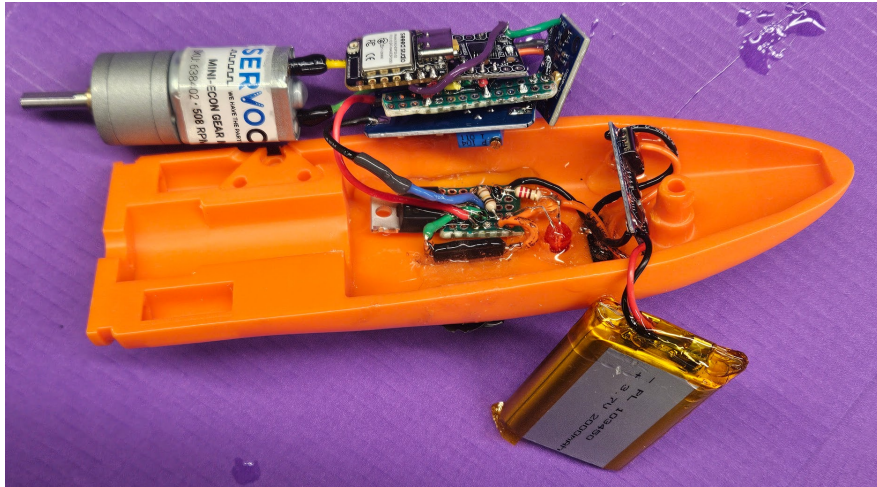


Figure 19: Complete circuitry built with breakout board modules.

Using the breakout modules (Fig. 19) allowed the circuit to be built on schedule, allowing for time to test performance and demonstrate the functionality of the upgraded RMO. With some careful soldering and a little finagling, everything fit snugly inside the fuselage, ready for testing. As shown above in Section 2.3, the fixed version of the PCB has since been delivered and functions perfectly. The very next step in this project is to rip open the RMO and replace those breakout modules with the updated PCB, which should improve performance.



Figure 20: The completed RMO build, ready for her maiden voyage

3 Performance Testing

Once the RMO was complete, the next step was examining how it performed. These performance tests were intended as a demonstration of functionality. The standard benchmark of the original RMO is a roughly 2Hz swimming gait at 5V. Speed decreases linearly with voltage supplied. At 2-3V, swimming gait typically drops to around 1Hz. The goal was to match these metrics, demonstrating equivalent performance with improved functionality.

3.1 Methodology

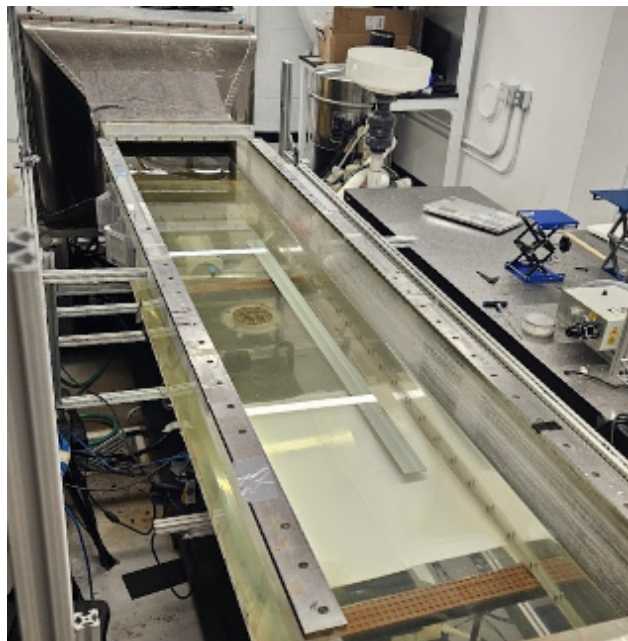


Figure 21: The water channel where the RMO is tested.

The scope of this Independent Work did not include any form of self-righting or steering

mechanism, which means the RMO still must be attached to some sort of support structure in order to swim straight. A 3D-printed tang was connected to an 18.5in wide square sheet of foam, as shown in Fig. 22. The intent of this was to hold the RMO straight inside the water channel (Fig. 21) as it swims, helping to standardize the motion for data collection. Unfortunately, the friction between the foam and the walls of the channel was too significant a factor, even with attempts to decrease friction, for the RMO to overcome. As the tail oscillated back and forth, the corners of the foam would be pressed into the glass, increasing the normal force and effectively halting the RMO's momentum after every tail beat.

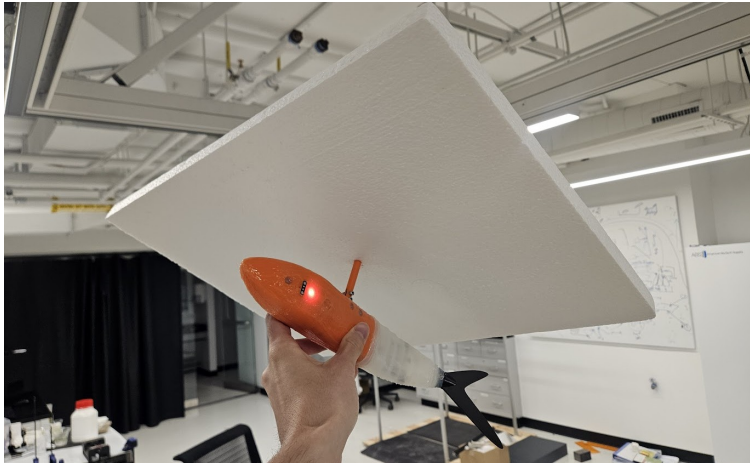


Figure 22: The RMO and its square foam mount.

To mitigate this, the foam square was trimmed down to 17in wide. This extra inch and a half of clearance allowed the RMO to "bounce" softly back and forth between the walls with every tail beat. Instead of slowing forward momentum, the normal force on the corners of the block just pushed the RMO back towards the other wall, because it had room to drift over. This definitely affected the performance metrics, because the RMO was wasting a small amount of energy moving back and forth instead of forward, but the RMO moved qualitatively much better than with the full size foam block. At higher swim speeds, the foam would smoothly bump both sides of the channel in succession before it could drift off center, leaving its path fairly linear, as is demonstrated in Fig. 23 below.

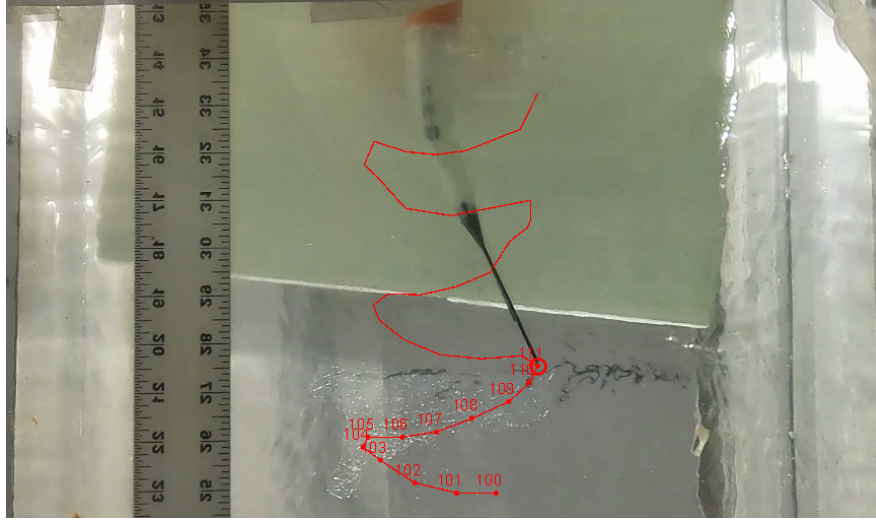


Figure 23: The underside view of the RMO swimming at 255 PWM (5V motor input). The path of the tip of the caudal fin was traced using Tracker and demonstrates its linear path of movement.

Without the capacity to record the positional data from the IMU while underwater (the RMO has no onboard memory installed – data can only be collected on a wirelessly connected device in the current setup), performance metrics had to be taken from recordings. Cameras were placed below and to the side of the water channel. See the camera views below in Fig. 24. The above view was blocked by the foam, as is shown in Fig. 25. Measuring sticks were attached to the channel so each camera could have a scale for RMO speed.

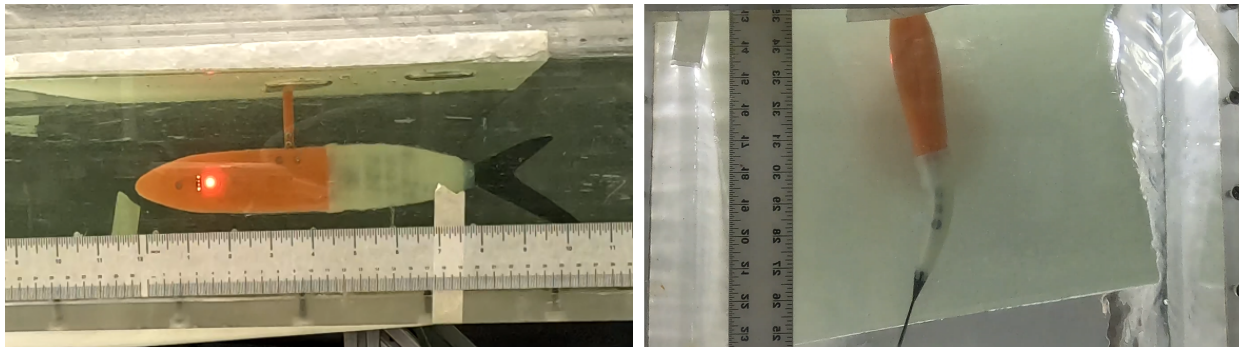


Figure 24: Side and below views of the RMO swimming.

The RMO was tested swimming linearly for a range of motor input PWM values from 255 to 210. Below 210, the motor was not strong enough to swim in water. PWM values of 255–210 are equivalent to driving the motor at voltages of 5V–4.1V. Five trials at each value in the range [210, 220, 235, 255] were performed. The RMO had to be regularly recharged

due to the unfortunate use of the MT3608 linear voltage regulator (Section 2.6), but overall it was not too much of an inconvenience.

Note: Since wireless connection was nonfunctional, it was not possible to change the swim speed of the RMO while it was submerged. However, when the motor is set to a certain speed before placing it into the water, the RMO will follow the last instruction received and keep swimming at that speed until the RMO is removed from the water, the signal reconnects, and the motor is turned off. This was the method that was employed to get trials at all the varying swimming speeds.

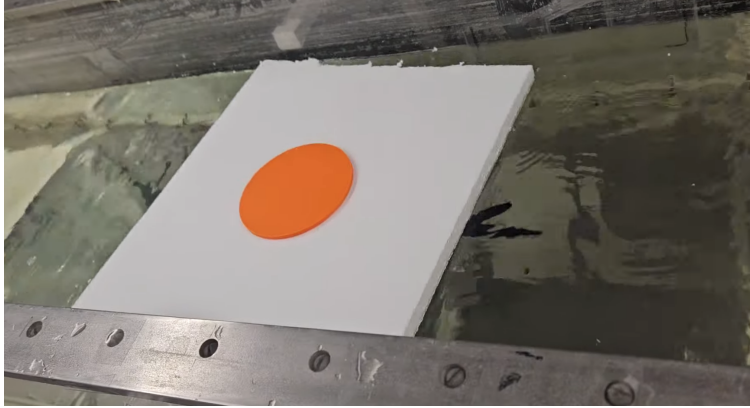


Figure 25: The RMO swimming, seen from above.

Once linear swimming trials were finished, the IMU had to be tested. Since wireless connection was nonfunctional, it was impossible to obtain the sensor data from the RMO while submerged (without breaking open the sealed fuselage to access the XIAO). To make sure there was still some form of demonstration, the IMU data was measured while the RMO was on dry land (Fig. 26). Two trials were performed, one with the motor off and one with the motor running, just to see how much the noise the motor introduces affects the data. For each trial, the RMO was lifted and then moved in all 6 DoF. This test showed the live positional data collection functions over wireless connection.

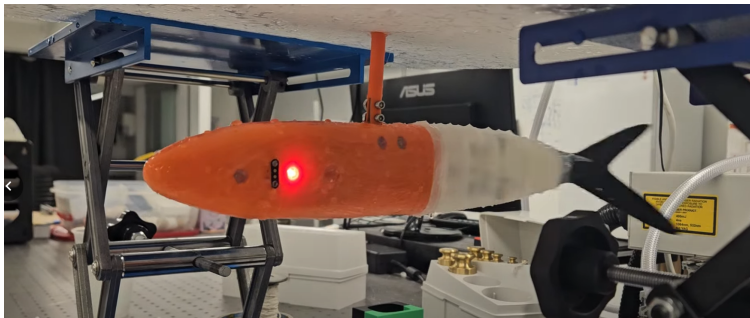
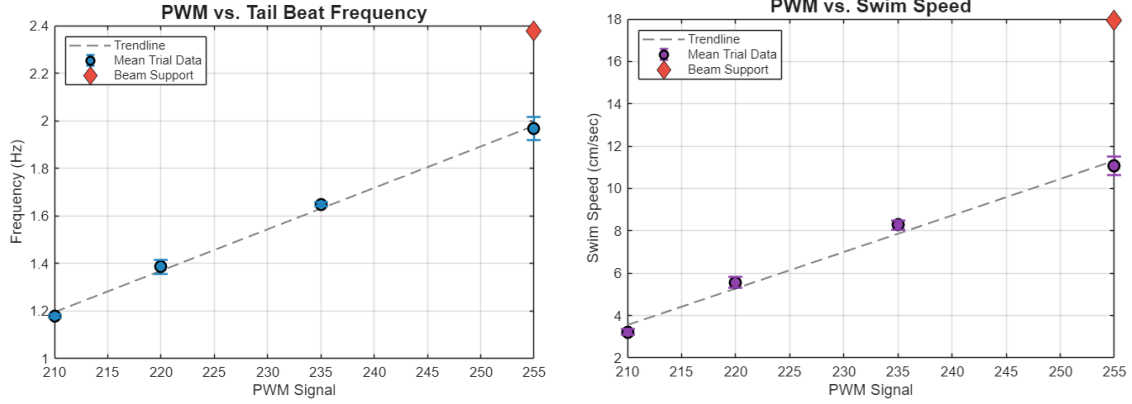


Figure 26: The RMO live-streaming IMU measurements.

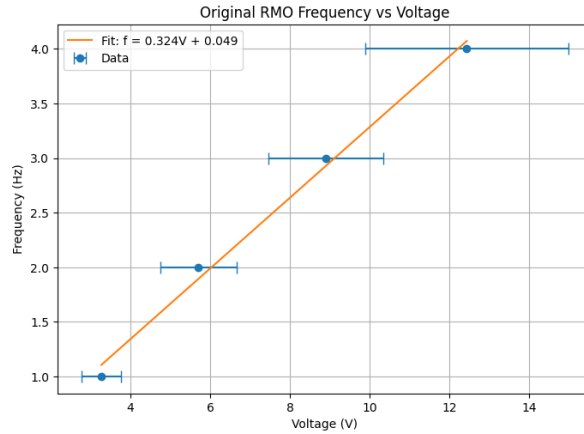
3.2 Results & Discussion

The camera recordings effectively captured the data, allowing for swim speed and frequency to be calculated to the millisecond with frame-by-frame manual analysis using a VLC Media Player plugin called Time v3.2. The RMO was reliable throughout the trials, although it was more finicky to set up than desired with the wireless connectivity issue and needing to be recharged.



(a) Demonstrates 2Hz at 5V and 1.2Hz at 4V

(b) Max speed 10cm/s



(c) Broad benchmark metrics set by original RMO.

Figure 27: Graphs demonstrating the performance metrics of the RMO when affixed to the square foam block support [Red point listed as "Beam Support" represents the RMO swimming straight when affixed to a small beam of foam]. A PWM value of 255 correlates to 5V input to the motor, and 210 correlates to approx. 4.1V. Note how fast the voltage-to-frequency ratio falls off as voltage decreases for this project's RMO compared to the original.

Fig. 27 shows that the RMO runs effectively at full speed, but rapidly loses power as the PWM signal drops, especially in comparison to the standard set by the original RMO. This is an interesting problem, indicating potential current supply issues at lower PWM levels.

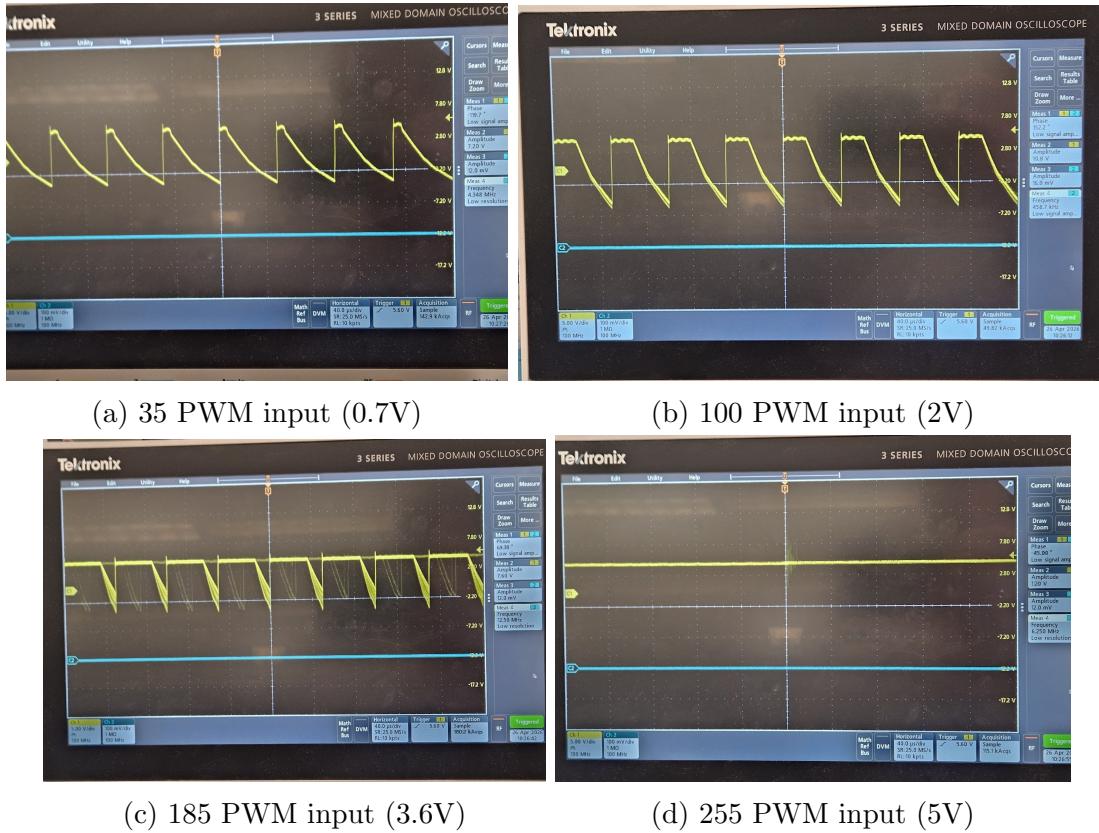
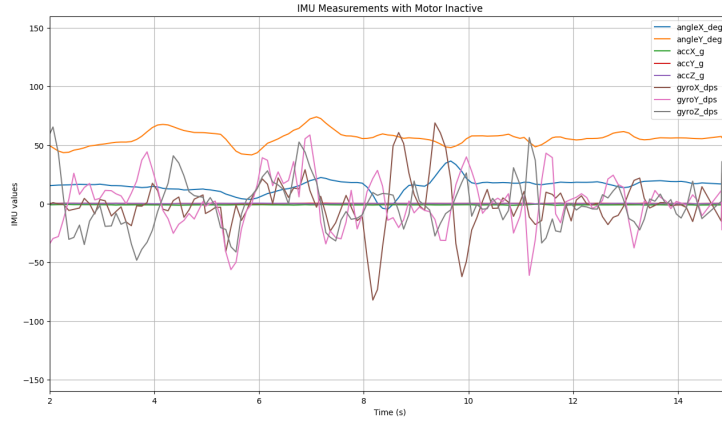


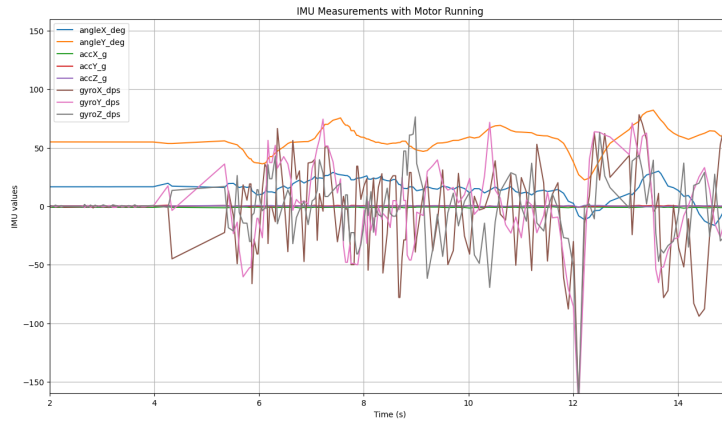
Figure 28: Oscilloscope readings across the motor terminals for the DRV8212 signal output at varying PWM values.

Oscilloscope readings of the motor signals at varying PWM values in Fig. 28 show interesting dynamics. Instead of a crisp square wave characteristic of a healthy PWM signal, the waveform exhibits RC decay (resistance-capacitance decay, where voltages drop in an exponential curve due to the resistive and capacitive values of the circuit). This could be the reason why the motor speed falls off so quickly. Voltage sag or insufficient power sourcing can prevent the full reset of current in the PWM signal, potentially resulting in reduced torque. More investigation into this needs to be performed. To debug the circuitry inside the RMO, it needs to be unsealed and cracked open, but it's important that this curious motor behavior be investigated further.

Dry land readings of the IMU shown in Fig. 29 demonstrate the functionality of the component. Notably, when the motor is on it introduces a ton of vibrational noise to the gyroscope. This makes sense but definitely hurts the IMU's usefulness as a data collection device. Some way of circumventing this should be identified moving forward.



(a) IMU Data being logged with Motor off



(b) IMU Data being logged with Motor on

Figure 29: These tests were taken while the RMO was on dry land. The IMU was tested via methodically moving the RMO through all 6 DoF. The motor was switched on at $t = 4$ in the second test, which can be seen in the large amount of gyroscopic noise introduced.

Angular data appeared nicely and showed resistance to motor noise. Very curiously, angle Z data didn't get recorded. This is very strange and potentially could be an issue with the MPU6050 chip used. Accelerometer data did get measured but is just scaled so much smaller than the others that it's not visible.

After data collection was complete, out of curiosity, the foam support block was removed. The RMO is still buoyant, which means it can awkwardly swim as it floats along the surface. Without support, it struggled to stay upright, although on a few notable occasions it flipped from its side to upright as it swam, staying stable for a few seconds before falling back over, as is documented in Fig. 30. We couldn't recreate it with any consistency, and it may just be chalked up to surface-fluid interactions between the fuselage of the RMO and the wall of the channel, but it was a cool behavior that's worth noting for a robot with no balance control system implemented.

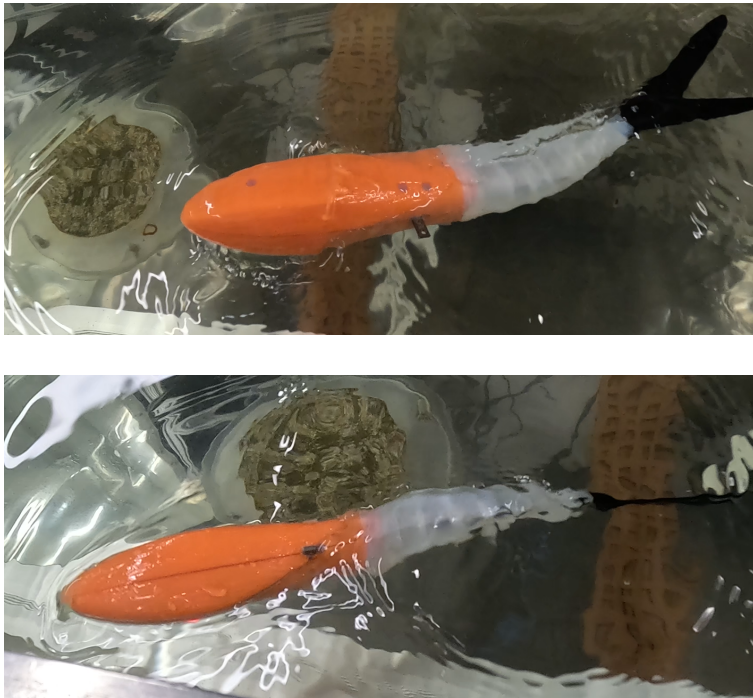


Figure 30: The RMO briefly self-rights as it swims unsupported.

After this, a foam beam was crudely attached to the RMO with fishing wire in lieu of the full foam block, just to investigate how it would swim, as can be seen in Fig. 31.



Figure 31: The RMO with beam support.

With the foam beam supporting the RMO, it was very stable upright. However, it tended to turn towards the left. Whether that is due to weight imbalances or misalignment of the foam beam is hard to say given the lack of controlled data. Sometimes though, the RMO would hit a good stride and travel for several feet in a straight line without hitting the walls of the channel, as was documented in Fig. 32. This was very inconsistent, but a recording of this was captured and used to inform the "Beam Support" data point in Fig. 27. Without the massive drag of the foam block, the RMO could move over 5 cm/s faster and achieved a 2.4Hz swimming gait, although this single measurement cannot be statistically significant.

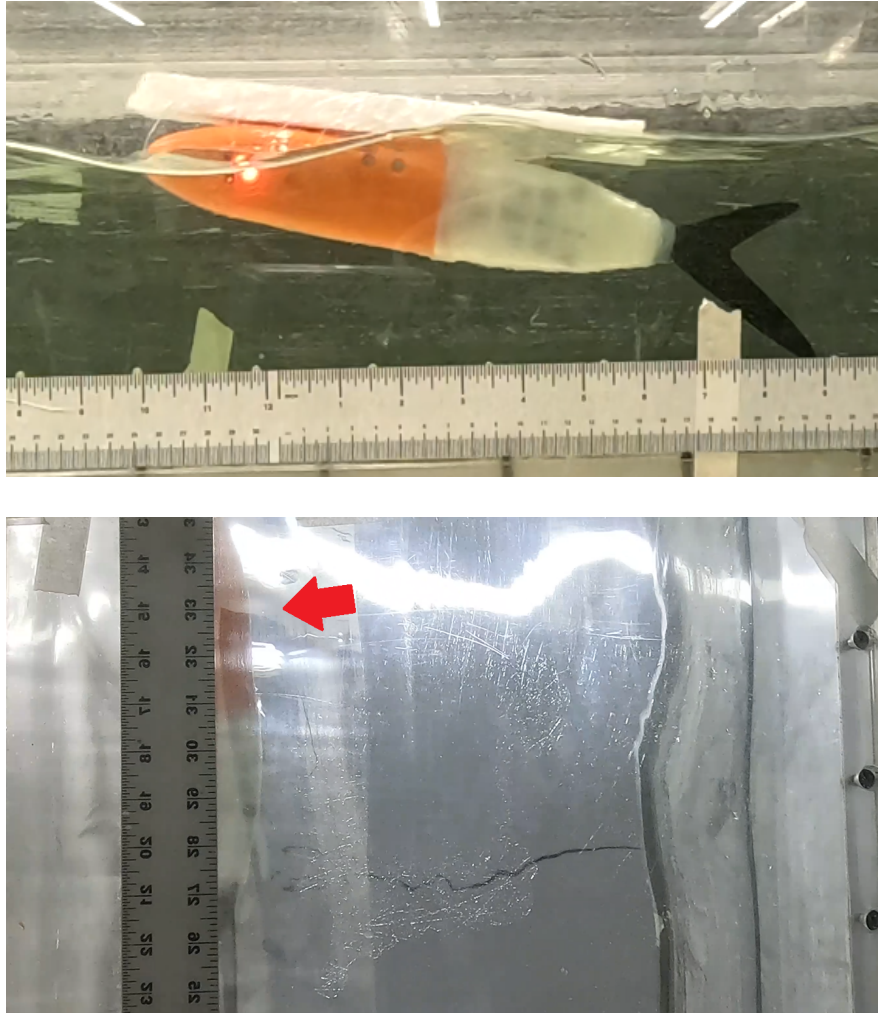


Figure 32: The RMO travels several feet along the length of the water channel, maintaining a 3-4in gap from the left wall.

4 Conclusion

This independent work project set out to improve an experimental tool. By upgrading the RMO into an electrical robotic platform, we become more capable to design and deploy studies which reveal deeper insights about the physics of flying fish. Through the integration of electrical power systems, an internal battery, sensors, and a programmable microcontroller, the RMO is primed to be a bio-relevant model in a wider array of applications than before.

Performance testing has demonstrated that the RMO can match the original RMO in swimming power at 5V. Thus, although it struggles to keep up as PWM values decrease (due to voltage sag and current drop from as-of-yet unidentified problematic wiring somewhere), we know it has the theoretical capability to match the original RMO at all voltages.

IMU data collection allows for accelerometer and gyroscope data to be recorded in real time, allowing for easy calculation of force and positional data. The WiFi connectivity issue poses a roadblock to the functionality of this feature, but as discussed below in Section 4.1, this is a solvable roadblock.

Qualitative swimming performances at high DoF have shown the RMO has some capacity to self-stabilize in certain indeterminate conditions, although that needs to be investigated further before any definitive statements can be made. The foam beam support is a promising option for developing an RMO with 4 or 5 DoF. The value of a biological model with that level of mobility is poignant.

The goal was to integrate electronics without negatively impacting the functionality of the RMO as an experimental tool. This was not fully achieved, as the current version of this project's RMO has some clear issues that hold it back compared to the original RMO. However, in other aspects, this project's RMO is already a better tool. As is always the case with experimentation, different jobs have different parameters. One of the biggest strengths of the upgrades added to the RMO are their versatility and controllability. Having a microcontroller brings so much control to the table, allowing for diverse behaviors to be directly programmed in. Better tools lead to more diverse experiments lead to a deeper understanding of both the biological and the engineered.

4.1 Future Steps

There are a wealth of next steps that can be taken to improve the RMO, starting with issues with the current version, then moving into new directions to explore with it.

The most obvious next step is to replace the breakout boards which are currently in the RMO (Section 2.6) with the functional second version of the PCB. This is an easy upgrade that's ready to go, and all the components are here and verified. But before we install any

new components into the RMO, we should examine the motor throttling at low PWM values.

The RMO should be opened up so that a lot of oscilloscope-ing can to be done on the motor driver board. Identifying where the voltage sag is coming from is critical to identifying what and where the issue is. One initial guess is that the bulk capacitance is too low, and increasing it from 100uf to 470uf may be an effective modification. Potentially as well, the PWM frequency is not optimal, being too high or too low at 15 kHz. Another option is that the current supply is being clamped too hard.

Additionally, the WiFi connectivity issue should be addressed. There are many ways to communicate wirelessly underwater [23], so switching to low frequency signaling may improve submerged range. The easiest solution, however, may be just including a small antennae that reaches up and out of the water.

Integrating Over-the-Air updating protocols would allow for the XIAO to be flashed new code wirelessly. This would be huge for the ease of use of the RMO, because as it stands the entire robot must be deconstructed to flash new software onto the microcontroller.

The RMO is naturally buoyant right now, which is kind of a problem for potential free swimming. Either manually tuning ballast or integrating some way to control buoyancy could really open up new DoF for the RMO.

Improving waterproofing methods is also a continuous pursuit. The current methods result in moisture accumulation inside the body cavity, which will wreck internal electronics over time even despite the conformal coating. Research into better methods of waterproofing would be valuable towards the robustness of the RMO.

The electrical platform itself opens many opportunities for increased experimental designs, forming a base for integration of future improvements, including sensor-based yaw control, electrical wing actuation, and modified swimming gaits. This project is super exciting and has many directions to go!

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5.1 AI Disclosure

AI tools were used to assist in the development of software used in this project. Particularly, AI tools helped build the HTML script that the ESP32 microcontroller uses to host the wireless RMO control website. AI tools were also used to assist in creating the scripts that generated the data figures in Matlab, as well as with LaTeX formatting.